

Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M2): Customer Development (CD) Model



For Tomorrow's Innovators

CIE Ignite Season02: A curriculum-integrated innovation program delivering guaranteed engagement with 4,500+ engineering students at their most motivated moment.

From classroom theory to real-world impact!

<https://cie.pes.edu/>



Glimpse of CIE Ignite Season 01 - 2025



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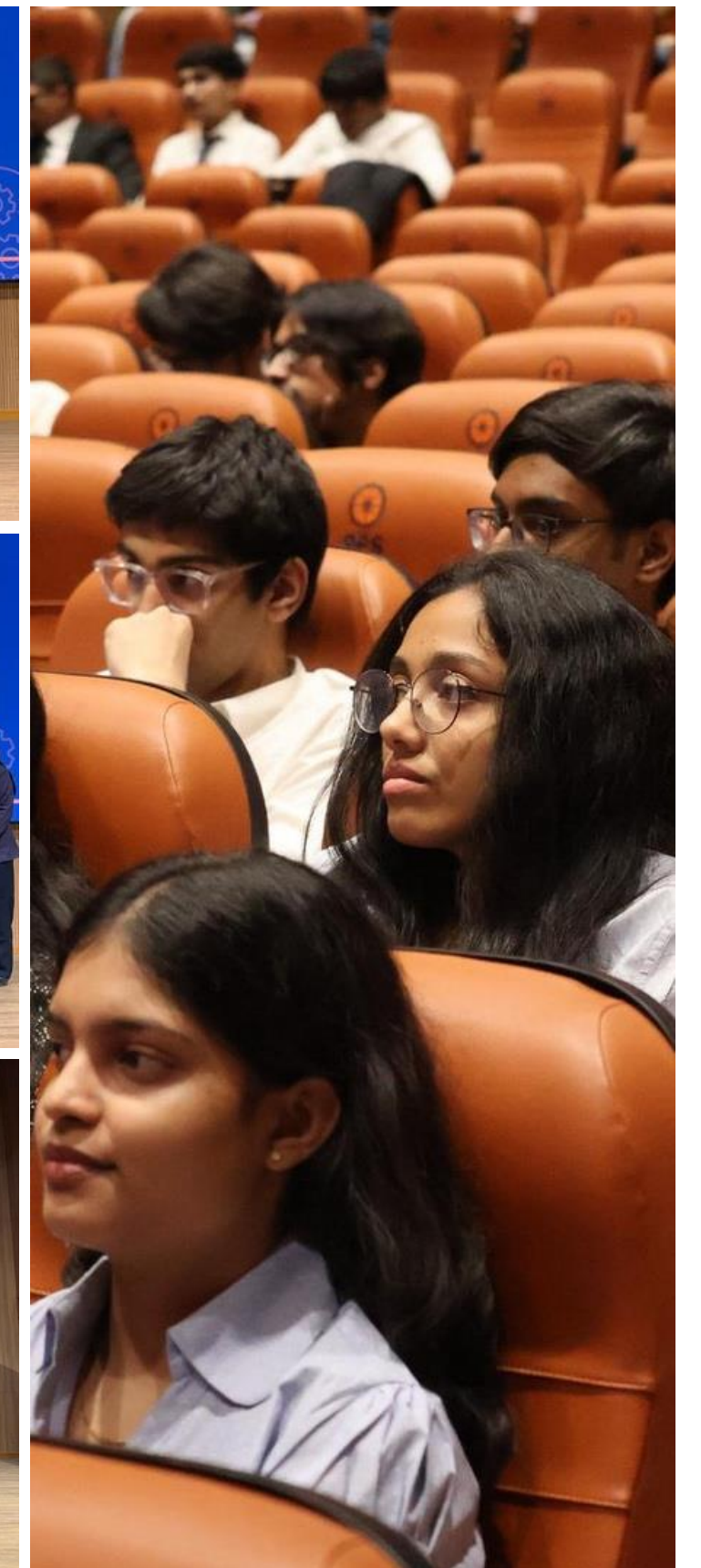
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Nurturing Excellence in Deep Tech



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CS, PESU Bangalore Chapter
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The CIE Ignite Journey



A three-phase innovation pipeline that transforms classroom learning into validated, market-ready solutions

Phase 1: Department Hackathons

January 2026

500+ teams compete across 21 departments. Top 20% advance with validated concepts and initial prototypes.

Phase 3: Grand Finale

March 2026

15 finalists pitch refined solutions to investors, partners, and faculty. Winners announced with prizes and opportunities.

Phase 2: Mentor Sprint

February 2026

60+ selected teams receive structured guidance from industry & alumni experts through two intensive mentorship sessions.

Beyond: Sustained Innovation

April 2026+

Top projects can continue as capstone work, enter pre-incubation programs, or launch as startups with ongoing support.

1

2

3

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The CIE Ignite Journey



Innovation starts here—and it does not stop.

Season 02 powers ideas into impact:

- Hackathons → Mentor Sprint → Grand Finale → Beyond
- Teams sharpen, validate, and pitch to real stakeholders
- Top projects continue as capstones, pre-incubation, or startups

Be part of the journey:

- Faculty: shape problem statements
- Industry: mentor, fund, and co-build
- Students: bring your best

Let us build market-ready solutions from PES!

Contact: tarunrama@pes.edu

Visit: <https://cie.pes.edu/>



Harvard Manage Mentor program

CIE – Harvard Manage Mentor 2026

“BEGIN YOUR JOURNEY FOR A SUCCESSFUL TRANSITION FROM UNIVERSITY TO CAREER”

Kickstart your career prep and unlock your full potential with Harvard Manage Mentor and more!



Bundle of 6 Topics from Harvard Manage Mentor

- Goal Setting
- Change Management
- Innovation and Creativity
- Strategic Thinking
- Sharpening your Business Acumen
- Presentation Skills



Course Structure

- [A] Asynchronous: Online Anytime, Anywhere !
- [B] In Person: Cafe Session in July/ Aug 2026



Course Certificate

Harvard Business Publishing Certificate
CIE course completion Certificate



Course Duration

Jan 2026 - July 2026

M1 Recap – Quiz

<https://www.menti.com>

Menti Code: 1335 3457



M1 Recap — Why Startups Fail

- Orientation & Course Overview
- Recall - What is a Startup?
- Why Startups Fail
- Traditional Business vs Lean Startup
- Lean Startup Principles
- Business Model Canvas
- Customer Development Process approach



Experiment – Google Classroom

- a. Received 40 Responses
- b. 19 Students with stu.pes.edu Email IDs
- c. 7 Students with pesu.pes.edu Email IDs
- d. stu.pes.edu Email IDs are not able to access Google Classroom. IT is looking into this issue.
- e. pesu.pes.edu Students can access, BUT only 4 of You accessed the Classroom
- f. 1 Student has completed the Assignment – Riya Halbhavi Thank You.
- g. We need more participants, We have reopened the Google Form.
- h. Please Volunteer for the Google Classroom Experiments
 - a. pesu.pes.edu will get access ASAP
 - b. stu.pes.edu please add your IDs and we will get you access soon

Experiment – Google Classroom



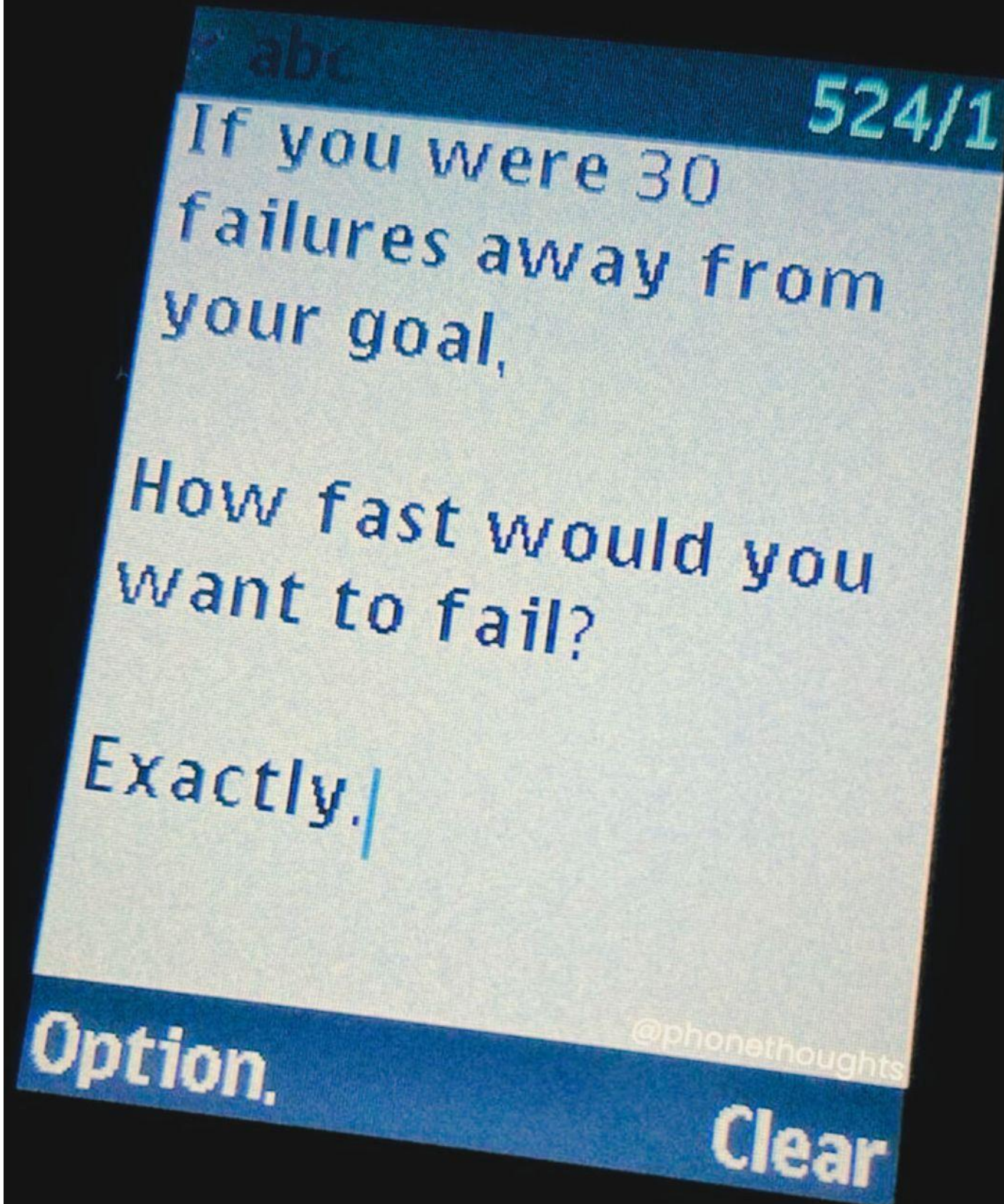
The #AI Corner

Top AI jobs

Founders

1	 Productivity/decision support	32.9%	
2	 Product ideation	19.6%	
3	 Company vision and strategy	19.1%	
4	Growth & GTM	9.3%	
5	Market analysis	8.0%	
6	Customer research	4.4%	
7	Team communication/alignment	1.8%	
8	Financial modeling/ops	1.8%	
9	Recruiting	1.3%	
10	Status reports	0.9%	

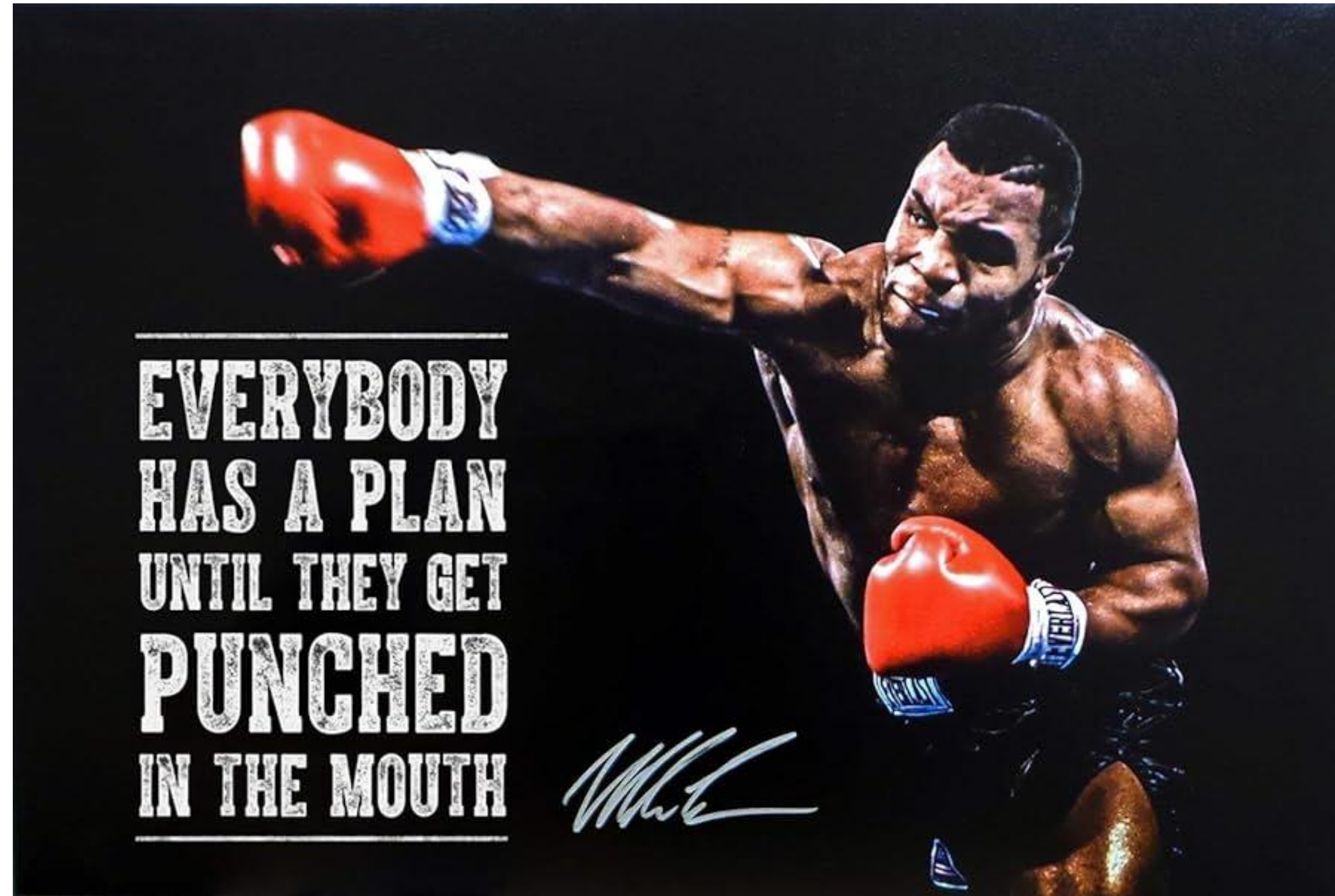
- Founders aren't using AI to *do work*. They're using it to **think better**.
- Their #1 use case is **decision support** — not slides, not code, not docs.
- Other roles treat AI as a tool. **Founders treat AI as a co-founder for judgment.**



Steve Blank, Eric Reis, modern Entrepreneurship

- **Steve Blank is a Silicon Valley entrepreneur and educator** who founded or worked in **8+ startups** across semiconductors, enterprise software, and consumer technology, giving him deep first-hand exposure to early-stage startup failures and successes.
- He is best known for creating the **Customer Development methodology**, which reframed startups as organizations that must **search for a repeatable and scalable business model**, not execute a fixed plan.
- Blank challenged the traditional “build first, sell later” approach by emphasizing **early hypothesis testing through direct customer interaction**—popularly phrased as *“get out of the building.”*
- His work directly influenced **Eric Ries and the Lean Startup movement**, with Ries explicitly crediting Customer Development as the conceptual foundation of Lean Startup practices.
- Leading organizations such as **Intuit and GE (FastWorks)** adopted Customer Development and Lean principles to reduce innovation risk and validate customer needs early, even at enterprise scale.
- The **U.S. National Science Foundation’s I-Corps program** is built on Steve Blank’s Customer Development model, demonstrating its global adoption beyond startups into research and academia.
- Today, his frameworks are widely taught in **universities, accelerators, and corporate innovation programs**, making Customer Discovery a core skill in modern entrepreneurship.

Blank said these (1/3) ...



Business plans rarely survive first contact with customers.

Blank said these (2/3) ...

5-Years Business Plan









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No one besides venture capitalists and the late Soviet Union requires five-year plans to forecast complete unknowns. These plans are generally fiction, and dreaming them up is almost always a waste of time.

Blank said these (3/3) ...

What is a startup?

“A startup is not a smaller version of a large company. It’s a temporary organization designed to search for a scalable business model.”

—STEVE BLANK

They do not unfold in accordance with master plans. The ones that ultimately succeed go quickly from failure to failure, all the while adapting, iterating on, and improving their initial ideas as they continually learn from customers.

Customer Discovery Overview

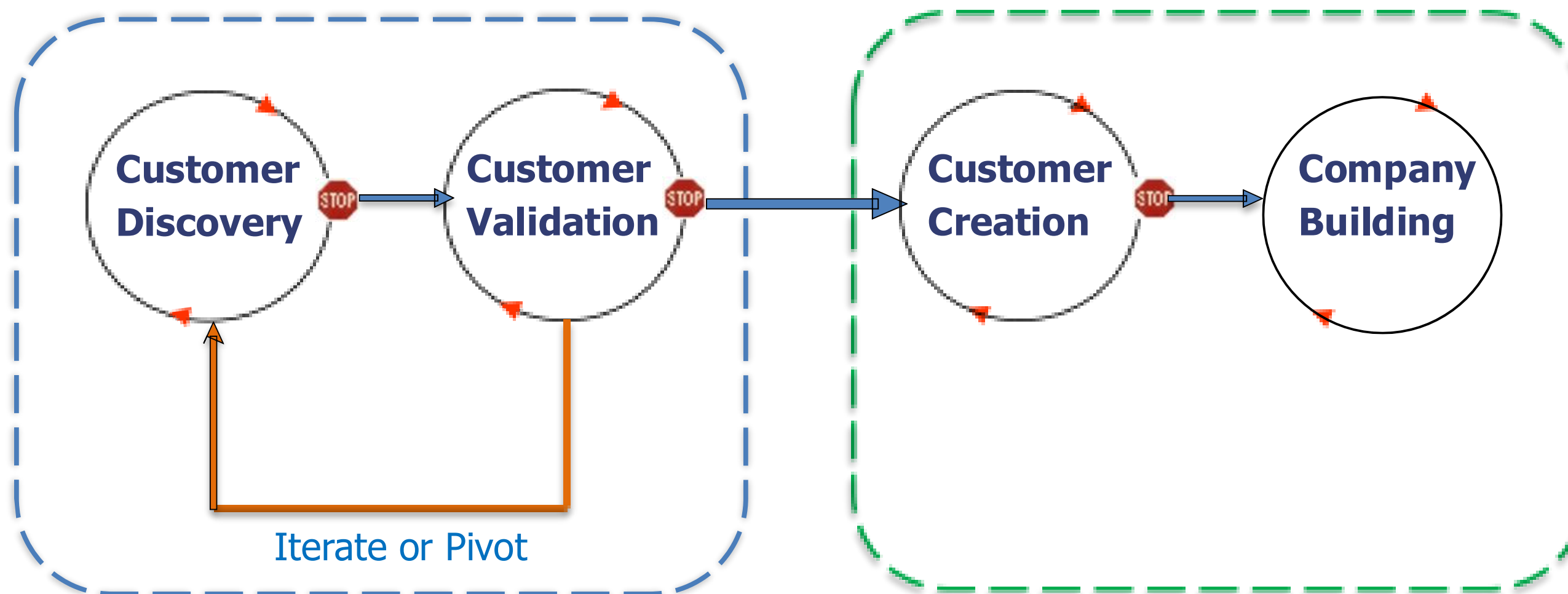
Customer Development



Customer Development Process

“Search”

“Execute”

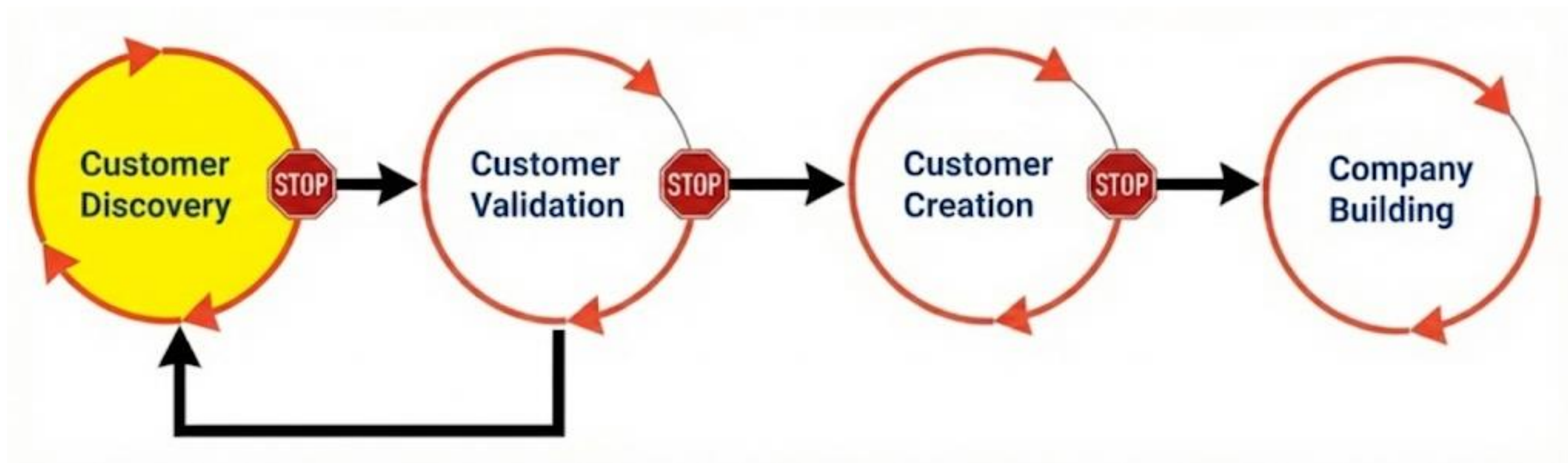


Source: Steve Blank

Video (2:41): [The Customer Development Process: 2 Minutes to See Why](#)

Customer Development

Step 1: Customer Discovery



STOP Selling, START Listening

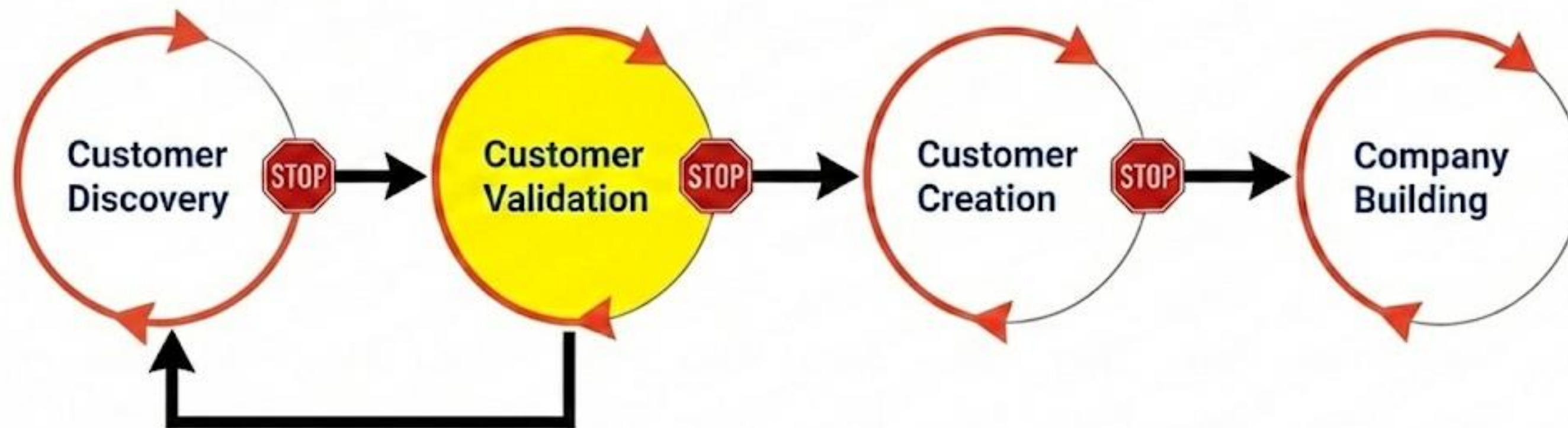
There are no facts inside your building, so **Get Outside of the Building (*GOOB*)!**

Test your hypotheses

Two main areas: **Problem** and **Solution** (can be either product &/or service) concept

Customer Development

Step 2: Customer Validation

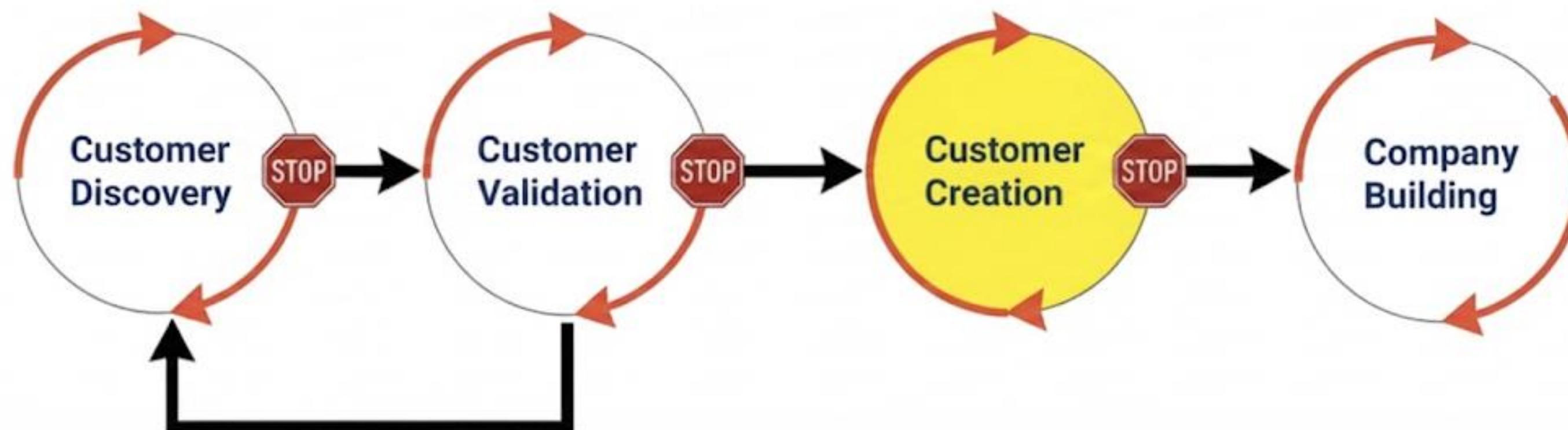


Validate your proposed solution against the Customer's problem(s)

- Is there a "**Product – Solution**" fit? "**Product – Market**" fit?
- Can you develop a repeatable sales process?
- Can you identify "**Early Adopters**" (open to trying out new/ semi-tested solutions)

Customer Development

Step 3: Customer Creation

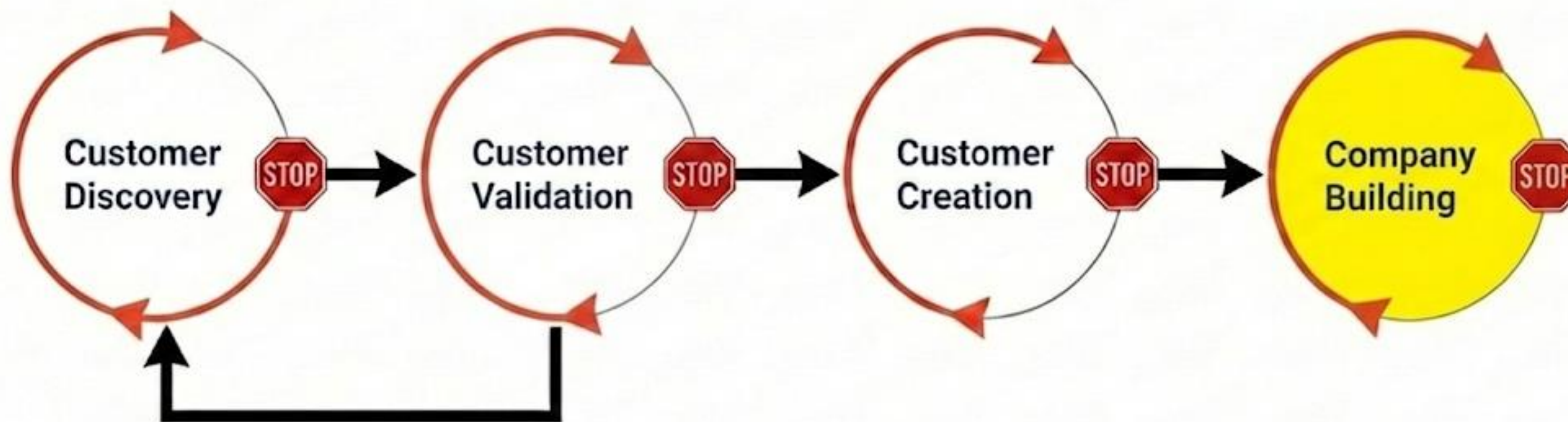


Creation (market/related) comes after proof of sales

- Can you execute your plan with speed and efficiency?
- Don't create something if you can't show sales!

Customer Development

Step 4: Company Building



Start the transition from Startup to a Company

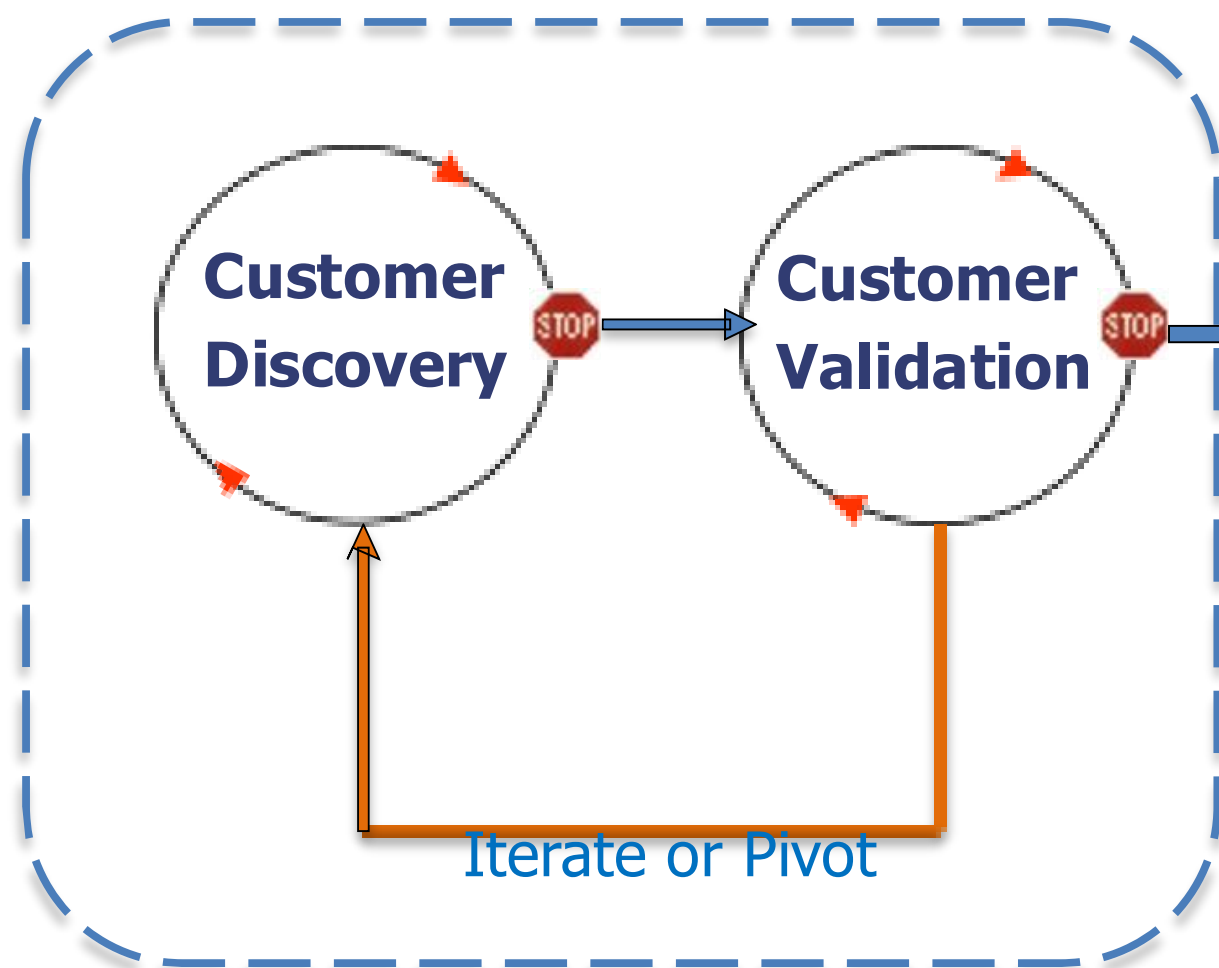
- (Re)build your company's organization & management....Why?
- Re-look at your mission (what got you here may not be sufficient)...Why?

Customer Development

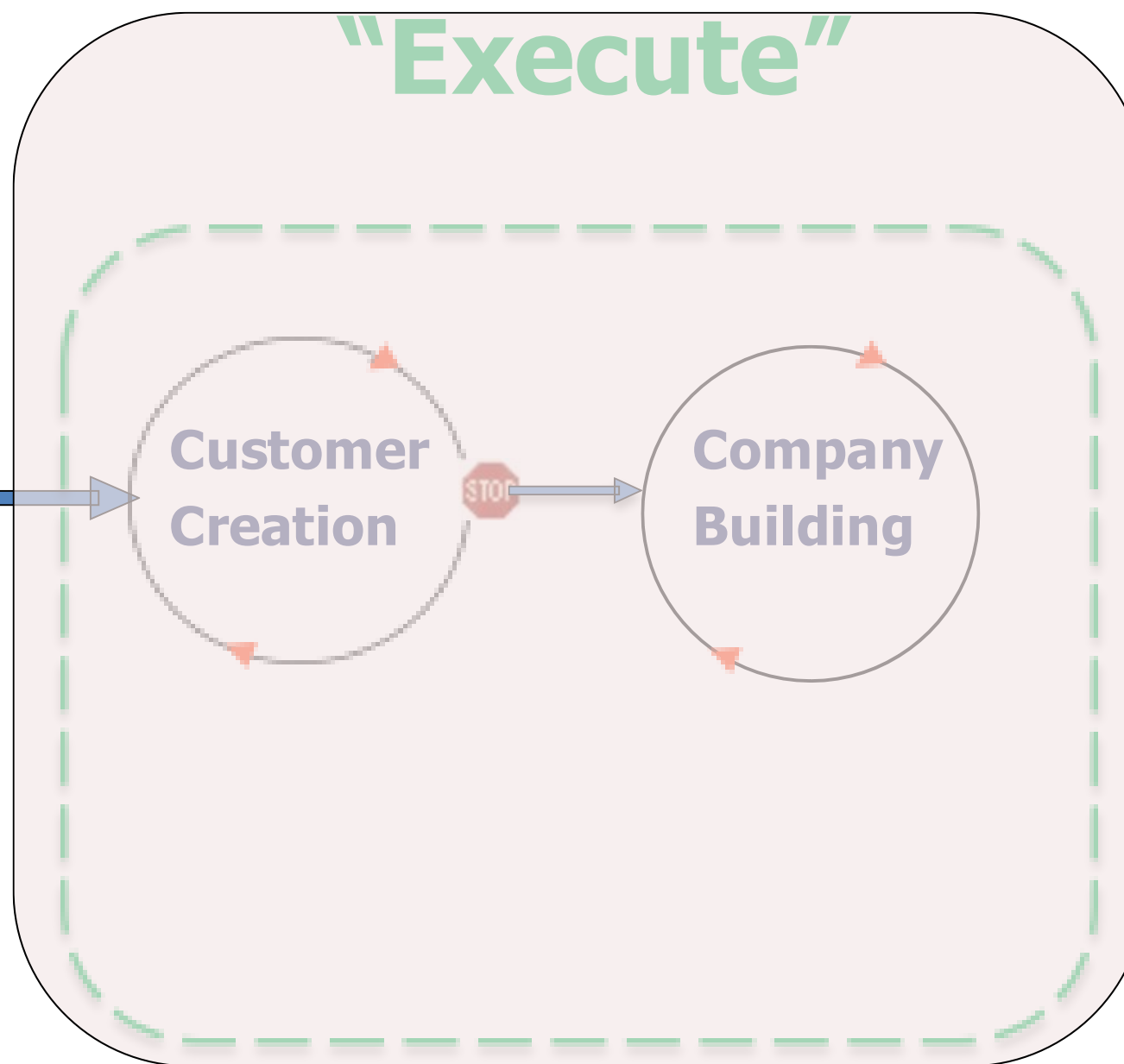


Focus Area for EIE2 & CIE-Ignite

"Search"



"Execute"



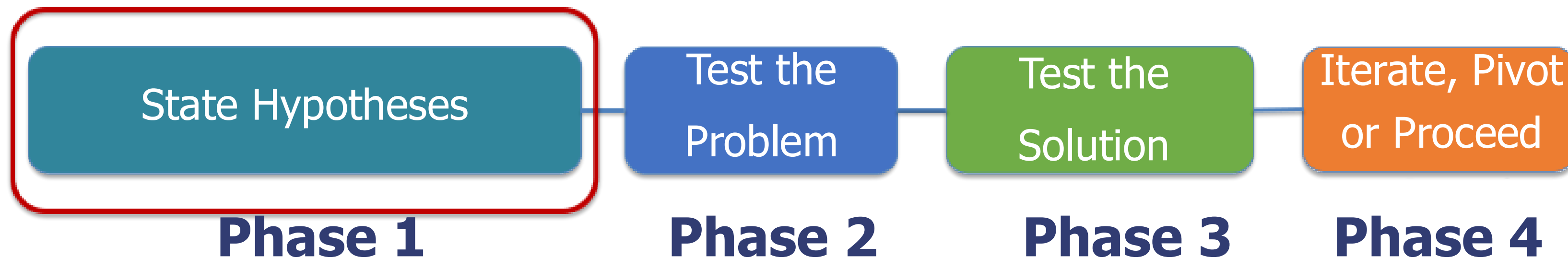
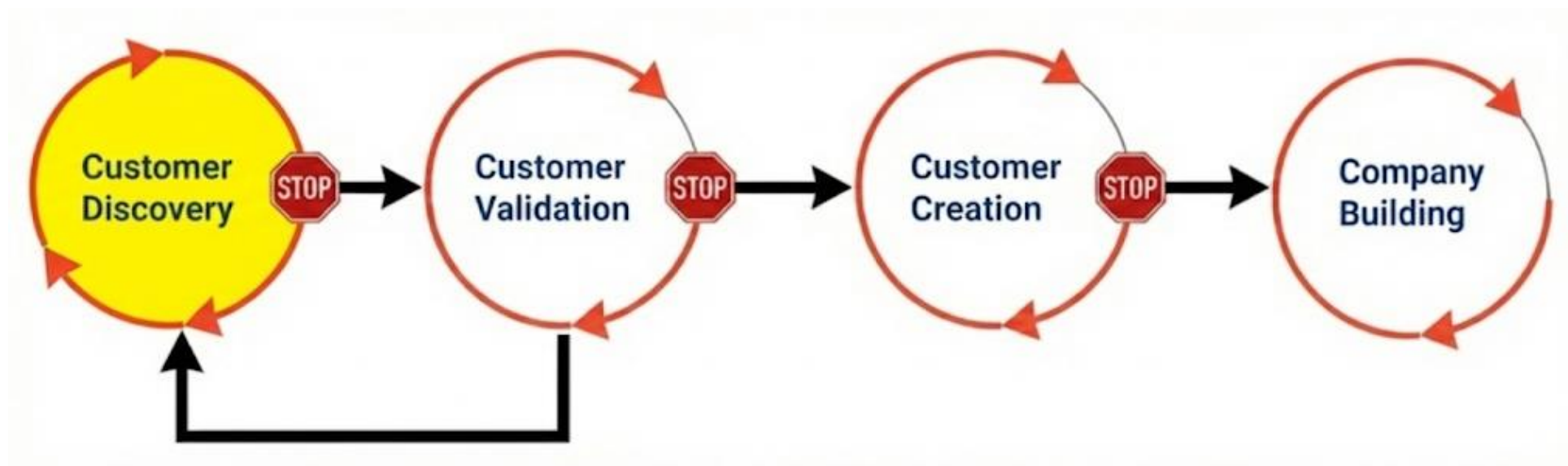
Hypothesis

- A **tentative, testable statement or assumption** made about a phenomenon, relationship, or outcome, which can be **validated or falsified through observation or experimentation**.
- It is formulated **before data is collected**, based on existing knowledge, reasoning, or assumptions, and serves to **guide systematic inquiry**.

Examples:

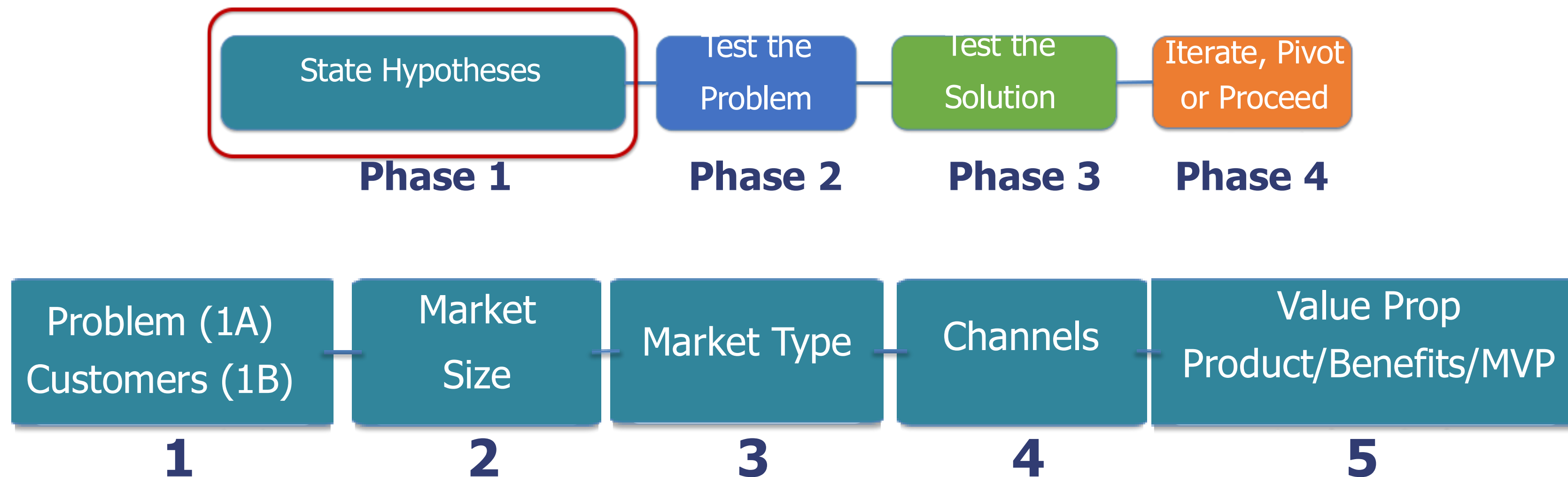
- Sounds Familiar:** "Students who supplement regular course material with 5–7 minute concept videos and audio will perform better in quizzes, ISAs and ESAs than students who rely only on live lectures and standard course material.": **Predicts a relationship between an intervention and an outcome**
- If the college canteen introduces UPI-only fast lanes during peak hours, average waiting time for students will reduce by at least 30%.: **Testable with a measurable outcome**
- "Urban Indian working professionals aged 22–35 prefer ordering groceries via mobile apps rather than visiting local kirana stores at least three times a week.": **Customer behaviour validated through data**
- "In T20 cricket, teams that score more than 50 runs in the powerplay have a higher probability of winning the match.": **Data driven assumption validated by match statistics**

Customer Discovery Overview



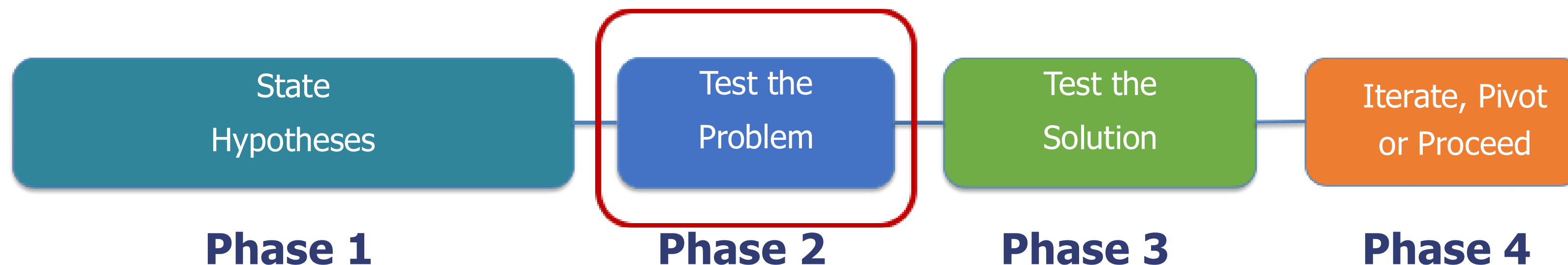
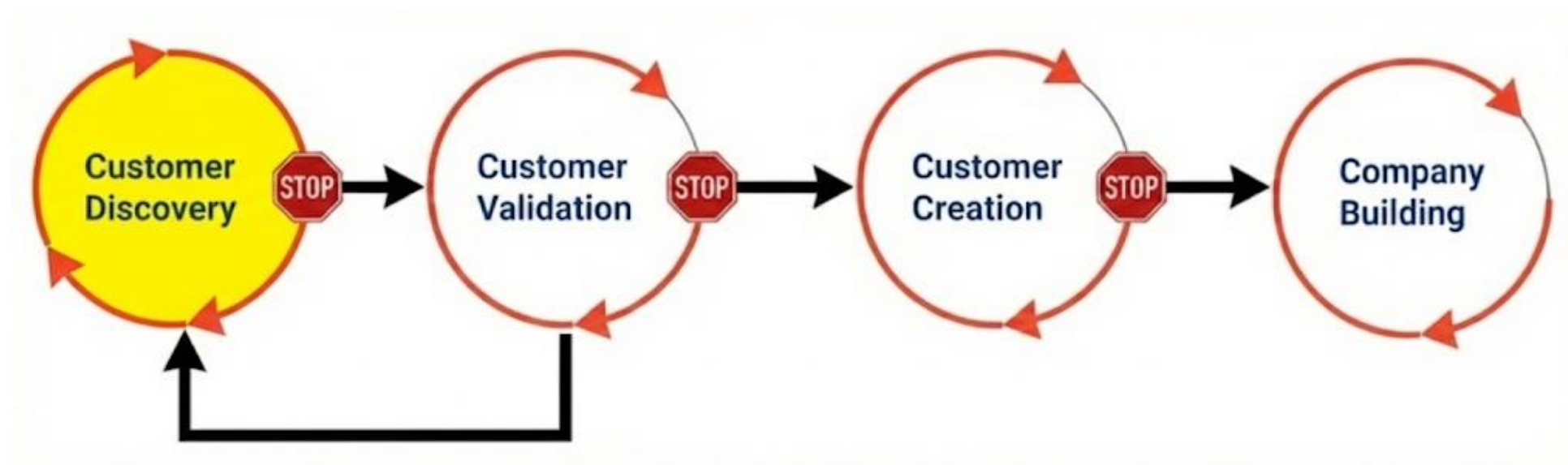
Customer Discovery Overview

Phase 1 : Hypothesis



**Capture your Hypothesis (Assumptions) on a
Business Model Canvas (BMC)**

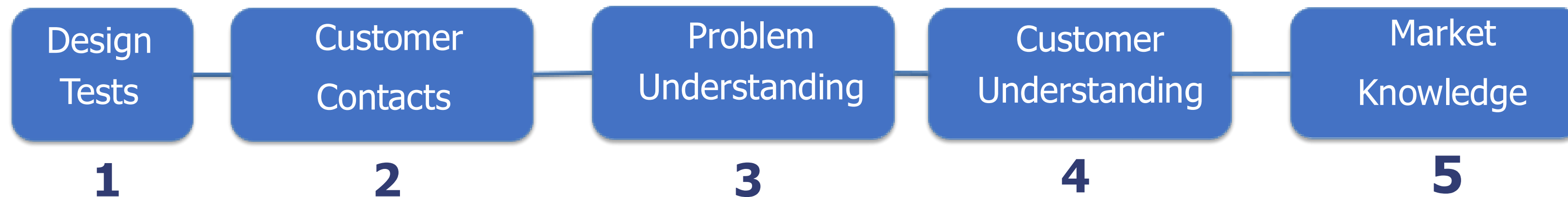
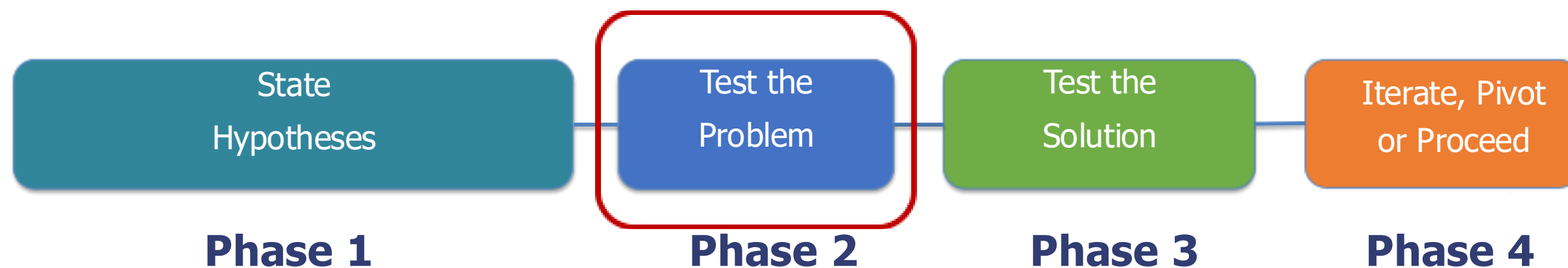
Customer Discovery Overview



Notice that testing the Problem comes before testing the Solution

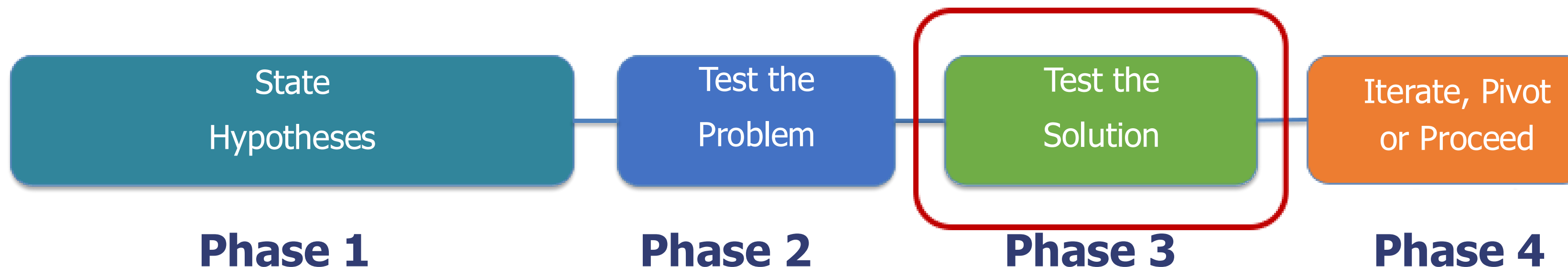
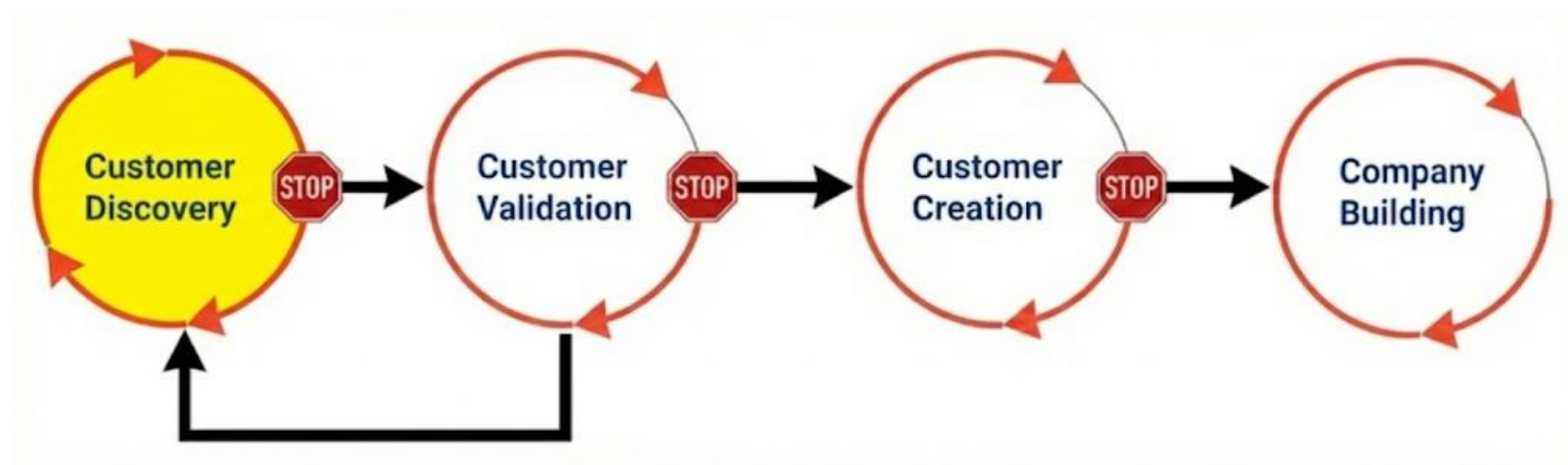
Customer Discovery Overview

Phase 2 : Test the Problem



Revise your BMC based on Testing your Problem

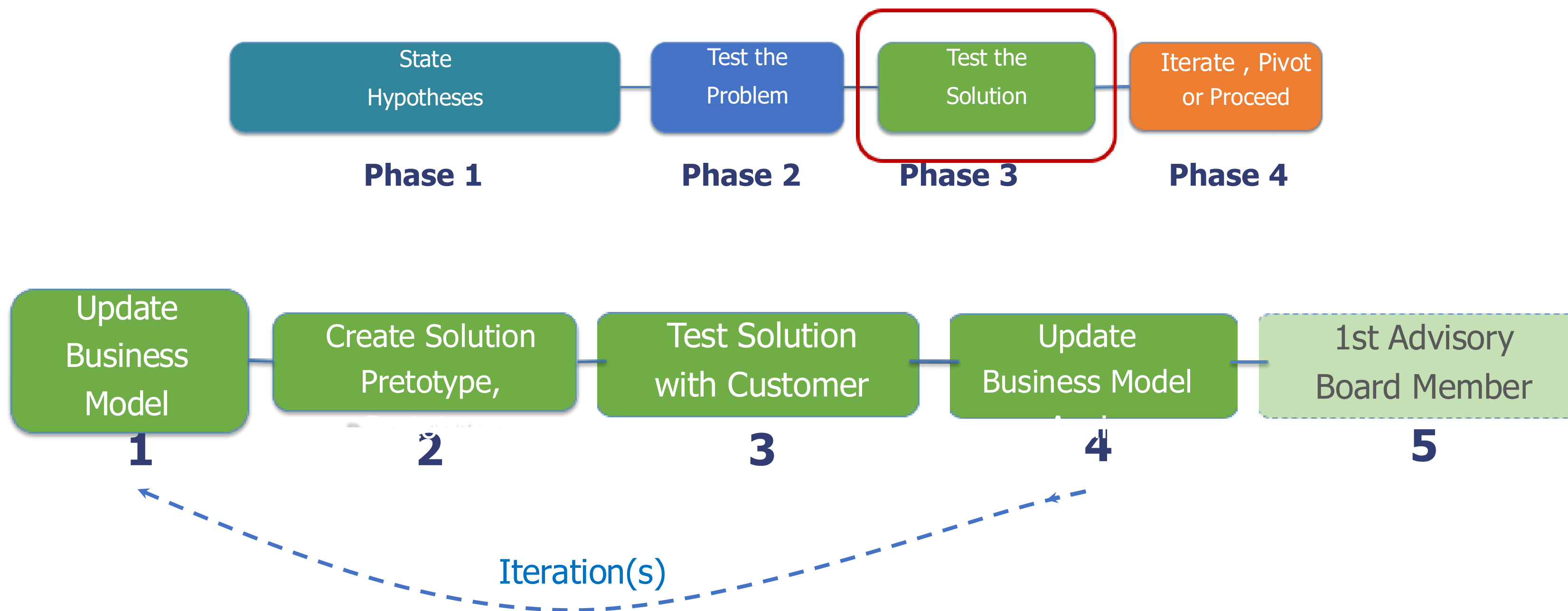
Customer Discovery Overview



Testing the **Solution** comes *after* narrowing down on the **Problem**

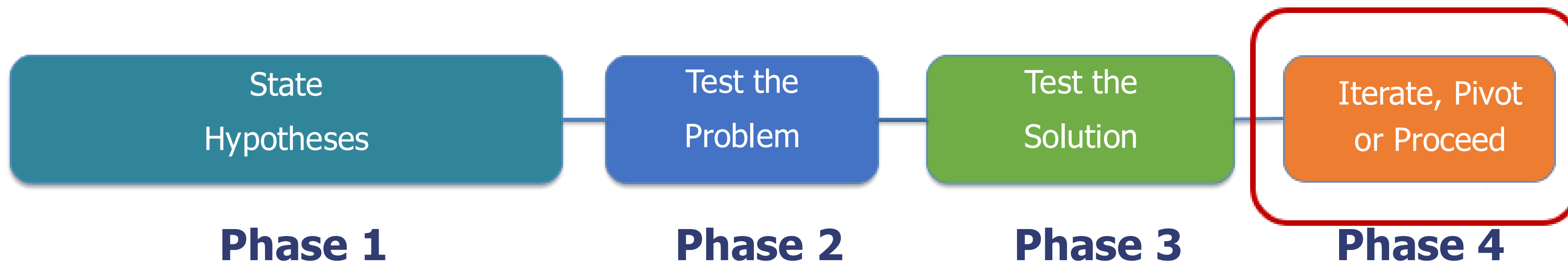
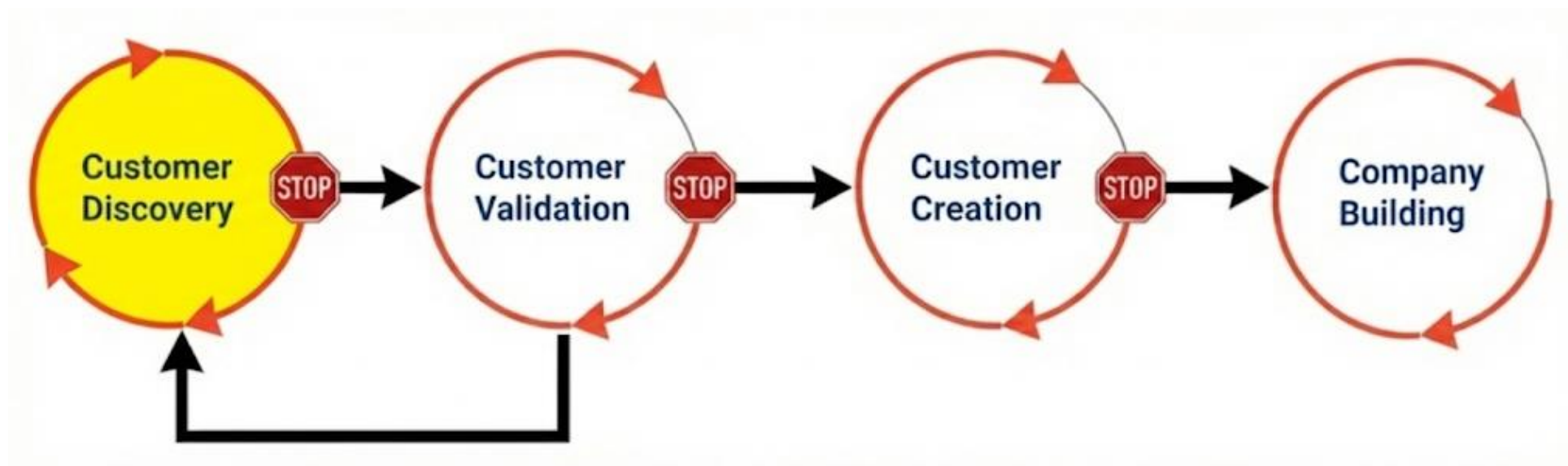
Customer Discovery Overview

Phase 3 : Test the Solution



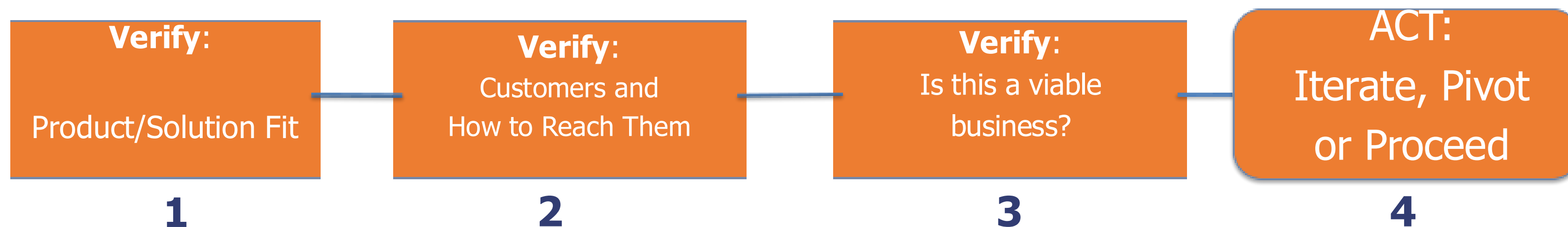
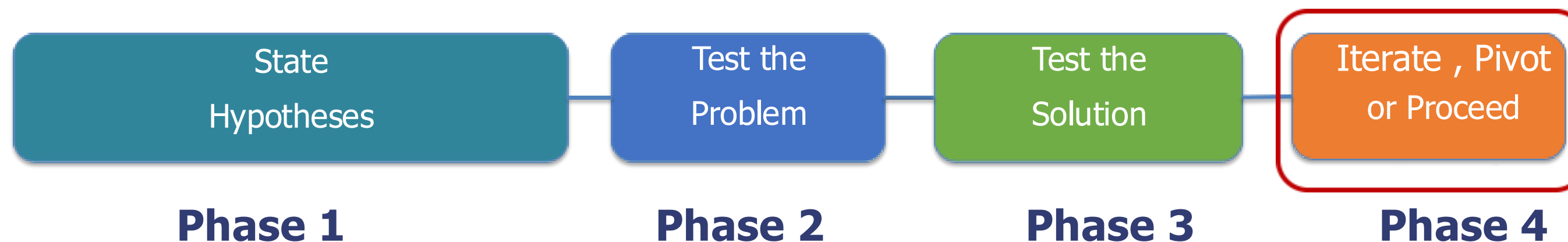
Iterate and Revise your BMC based on Testing your Solution

Customer Discovery Overview



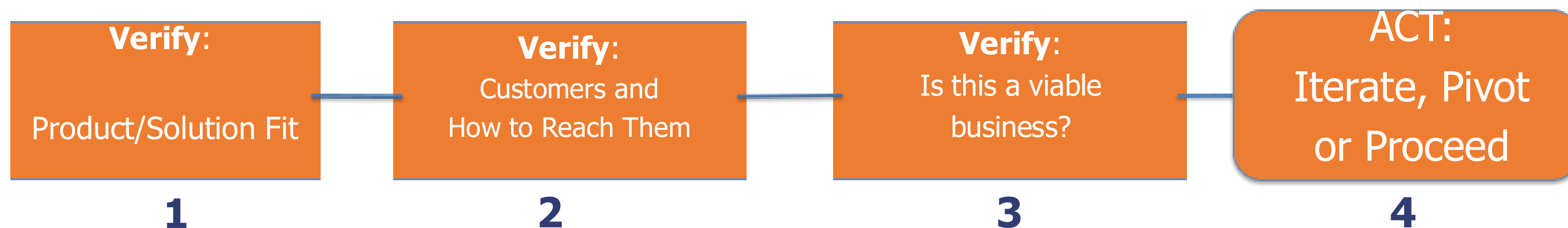
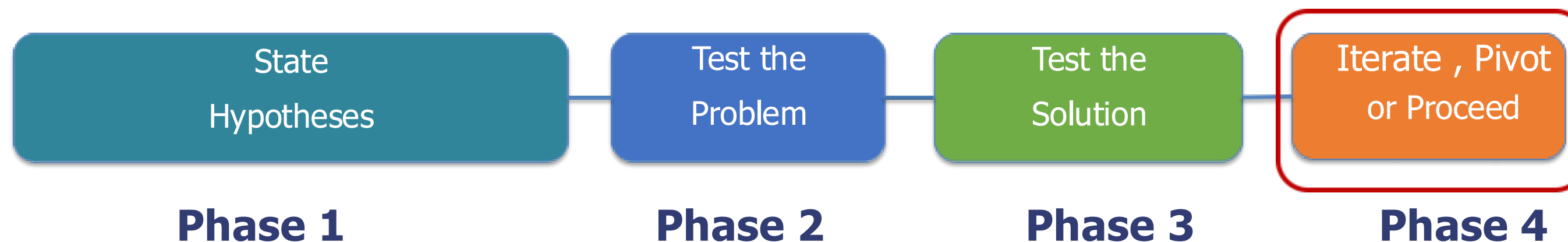
Customer Discovery Overview

Phase 4 : Iterate, Pivot or Proceed



Customer Discovery Overview

Phase 4 : Iterate, Pivot or Proceed



Build-Pivot-Kill-Buy

Classroom Decision Scenario

Product Name (fictional): *NoteSync AI*

An AI tool that automatically records, transcribes, summarizes, and organizes college lectures into searchable notes for students.

Target Users:

- Undergraduate students (18–22 years)
- Online / hybrid learners

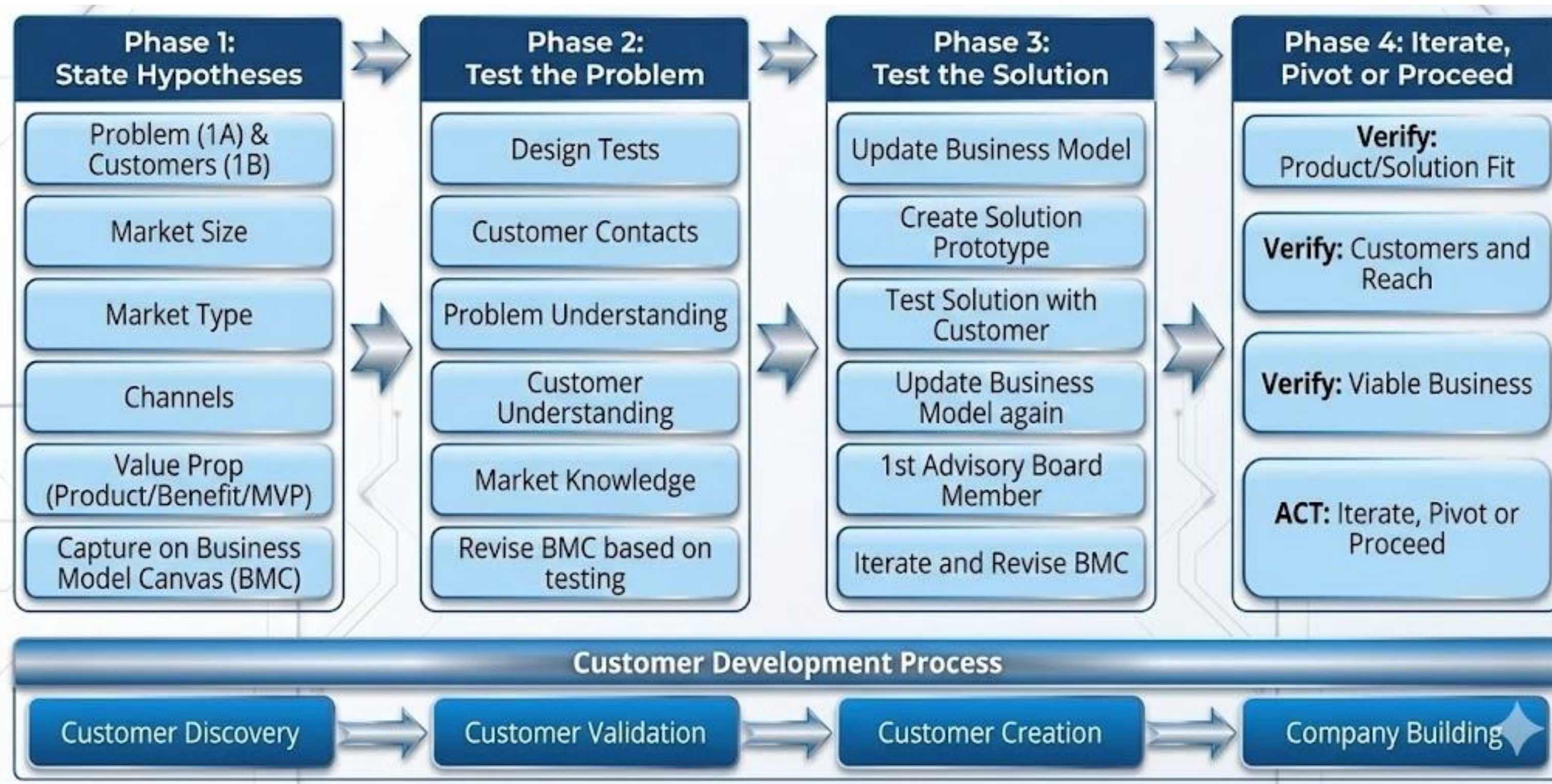
Core Features:

- Auto lecture recording
- AI-generated summaries
- Keyword-based search
- Integration with Google Classroom / Teams

Business Model:

- ₹299/month student subscription
- ₹99/month student subscription
- Sell B2B Licences to Universities/Colleges
- Freemium with limited summaries

Customer Discovery Overview



'9 Deadly Startup Sins' by Steve Blank

[Link](#)

Assuming “I know what the customer wants”

The “I know what features to build” flaw

Focus on launch date

Emphasis on execution instead of hypotheses, testing, learning and iteration

Tradition business plans presume no trial and no errors

Confusing traditional job titles with what a startup needs to accomplish

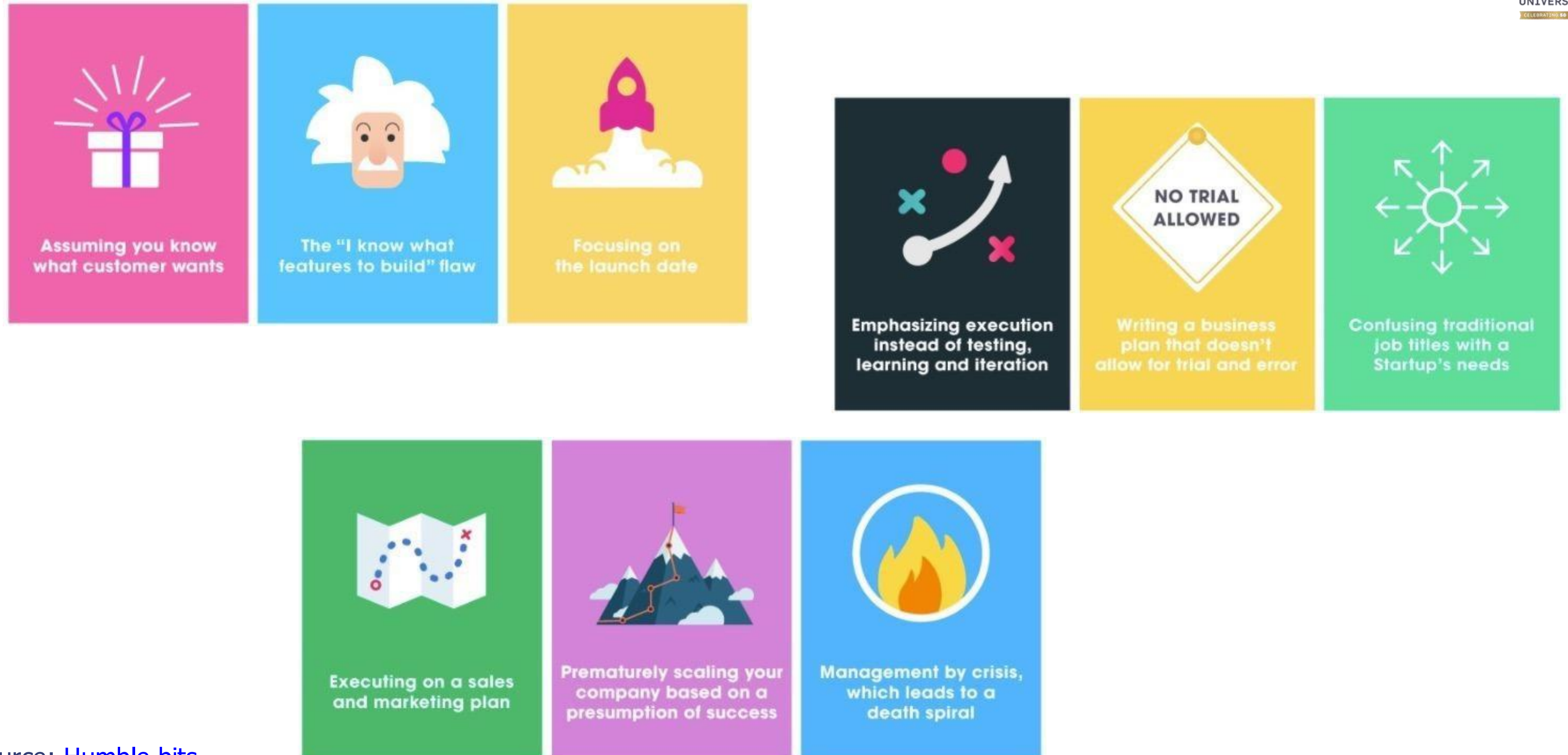
Sales and Marketing execute to a plan

Presumption of success leads to premature scaling

Management by Crisis leads to “Death Spiral”

System Preferences

'9 Deadly Startup Sins'



Examples



Dropbox

Customer Discovery: Drew Houston, the founder of Dropbox, initially created a video demo of the product and posted it on a technology forum to gauge interest and gather feedback. The overwhelmingly positive response from potential customers helped validate the idea and demonstrate the demand for an easy-to-use file storage and sharing solution.

Impact: Armed with valuable customer feedback, Dropbox was launched in 2007, offering cloud-based file storage and synchronisation. The company's customer-centric approach, which focused on addressing user needs and pain points, contributed to its rapid growth and widespread adoption. Today, Dropbox is a leading cloud storage provider with millions of users worldwide.

Examples



Zappos

Customer Discovery: Zappos, an online shoe and clothing retailer, has built its success on a foundation of exceptional customer service. In the early stages, founder Tony Hsieh recognized the importance of understanding customer needs and expectations. He personally spoke with customers to gain insights into their preferences and pain points.

Impact: Zappos prioritised customer satisfaction and went beyond selling products by focusing on creating a delightful shopping experience. This customer-centric approach paid off, and Zappos became known for its outstanding customer service and loyalty. In 2009, Amazon acquired Zappos for over \$1 billion, recognising the company's commitment to customer satisfaction.

Summary

- **Outside Perspective:** Facts lie beyond your walls.
- **Problem-Solution Testing:** Hypotheses validation.
- **Product-Market Alignment:** Early adopter validation.
- **Sales as Proof:** Create after validating sales.
- **Transition Wisely:** Evolve from startup to company.
- **Adaptive Growth:** Embrace chaos for innovation.



Expert Talk: Medha Koppam



Expert Talk: Medha Koppam



Current Role

Medha is a Research Fellow at Microsoft Research, focused on AI for Software Engineering and building reliable tools for large-scale code changes. She works on shipping features used across teams and co-designs AI systems for code transformation and testing.

Previous Experience

She previously interned at Couchbase and Rakuten, working on AI-powered optimization and debugging.

Leadership & Recognition

She served as the British Deputy High Commissioner for a day.

Achievements & Competitions

Under CIE's mentorship, her team won the Cisco ThingQbator contest for a mental-health app idea.



Expert Talk: Rahul Srinivas



Expert Talk: Rahul Srinivas



Current Role

Rahul Srinivas is Chief of Staff at AstraFin Neosolutions, working from the Founder's Office across product, go-to-market, and strategy. He drives product execution, GTM, and cross-functional delivery.

Previous Experience

He was previously Founder's Staff at Klub in Singapore, leading fundraising, investor relations, strategy, and new products. Earlier, he worked in the SaaS Pay Founder's Office from Associate Product Manager to Senior Manager across product, partnerships, and revenue.

Entrepreneurship & Industry Experience

He co-founded Sublit with over ten thousand daily users, with technology adopted by PES University, taught at Refining Skills, and interned at Bosch and Liebherr on data-driven manufacturing and analytics.



Expert Talk: Kartikeya Jain



Expert Talk: Kartikeya Jain

Education: B.Tech in Computer Science & Engineering, PES University (2021)



Current Role: Technical Project Manager at Amadeus, driving end-to-end planning, execution, and risk mitigation for customer-facing services and incident management.

Innovation & GenAI exposure: Holder of two international intellectual property disclosures (Defensive Publications) in the field of Generative AI, focused on building internal efficiency tools.

Career Path: Progressed from Developer → Customer Delivery Manager → Scrum Master → Program Manager, gaining comprehensive exposure to both technical execution and business portfolios.



Team Activity 1 - Know Your Team

Members

1. Partition a KG cardboard* area into (# of members + 1) sections
 2. For each member, capture the response to below questions in respective sections:
 - a. What best describes you? (1 word/sentence)
 - b. One thing your team should know about you to get along well.
 - c. Who's your role model?
 - d. Happiest moment of your life.
 - e. Most difficult moment/incident of your life.
 - f. One game-changing (even if 'impossible') idea
 1. Fill the remaining section, capture 10 things that you all/team members have in common
- * This can be adapted for virtual environment – be creative!

Team Activity 2 – Your Team Identity / Introduction

- Team Name/Logo, Members Name & Photo, Tag-line
- Team Member role(s) in the new startup
- 1 Fun Fact about the team 😊
- Team Goal (what you plan to create/accomplish by semester end)
- What makes us a Team:
 - Strength, Weakness
 - Common interests, values, beliefs, etc
 - 2-3 areas/themes team is excited to explore (e.g health, edu, etc)



1 Slide (see backup for template)
(becomes team's cover slide for *all* future presentations; get creative in designing this!)

1-2 slides (or video) –
Get creative!

Report out
[5 mins/team]